

Exploring the Vulnerabilities of Federated Learning

: A Deep Dive into Gradient Inversion Attacks

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백 연우

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연합학습과
관련되어 있는가



최근에 출판된
논문인가



구현체가
존재하는가

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시대적 흐름

개인정보 공유 없이도 학습한다...병원·은행서 통하는 ‘연합학습 AI’ 개발

데이터 공유 없이 고성능 AI 실현...KAIST, 연합학습 기술 개발

AI 발전 속 개인정보 침해, 연합학습으로 막는다

논문 소개

논문 소개

제목 (영문) :

Exploring the Vulnerabilities of Federated Learning
: A Deep Dive into Gradient Inversion Attacks

제목 (국문) :

연합학습의 보안 취약성 연구: 그래디언트 역전 공격을 중심으로

논문 저자 :

Pengxin Guo, Runxi Wang 외 8명

논문 출처 :

Hugging face - papers

Exploring the Vulnerabilities of Federated Learning: A Deep Dive into Gradient Inversion Attacks

Pengxin Guo*, Runxi Wang*, Shuang Zeng, Jinjing Zhu, Haoning Jiang, Yanran Wang, Yuyin Zhou, Feifei Wang, Hui Xiong, Fellow, IEEE, and Liangqiong Qu
Project page: [pengxin-guo.github.io/FLPrivacy](https://github.io/FLPrivacy)

Abstract—Federated Learning (FL) has emerged as a promising privacy-preserving collaborative model training paradigm without sharing raw data. However, recent studies have revealed that private information can still be leaked through shared gradient information and attacked by Gradient Inversion Attacks (GIA). While many GIA methods have been proposed, a detailed analysis, evaluation, and summary of these methods are still lacking. Although various survey papers summarize existing privacy attacks in FL, few studies have conducted extensive experiments to unveil the effectiveness of GIA and their associated limiting factors in this context. To fill this gap, we first undertake a systematic review of GIA and categorize existing methods into three types, i.e., *optimization-based* GIA (OP-GIA), *generation-based* GIA (GEN-GIA), and *analytics-based* GIA (ANA-GIA). Then, we comprehensively analyze and evaluate the three types of GIA in FL, providing insights into the factors that influence their performance, practicality, and potential threats. Our findings indicate that OP-GIA is the most practical attack setting despite its unsatisfactory performance, while GEN-GIA has many dependencies and ANA-GIA is easily detectable, making them both impractical. Finally, we offer a three-stage defense pipeline to users when designing FL frameworks and protocols for better privacy protection and share some future research directions from the perspectives of attackers and defenders that we believe should be pursued. We hope that our study can help researchers design more robust FL frameworks to defend against these attacks.

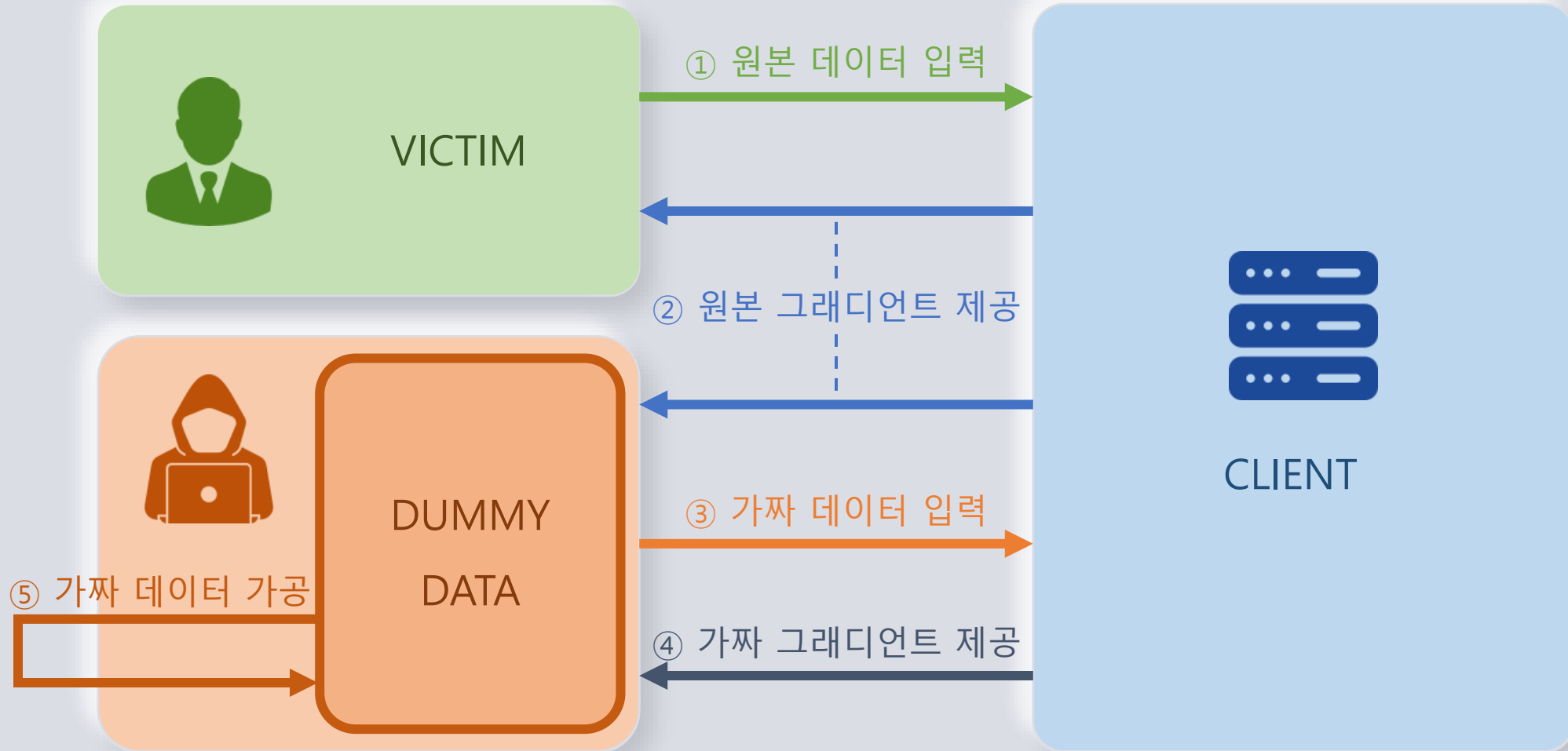
Index Terms—Federated Learning, Data Privacy, Gradient Inversion Attacks.

I. INTRODUCTION

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Feifei Wang is with the Department of Electrical and Electronic Engineering, The University of Hong Kong, Hong Kong 999077, China, and also with the Materials Innovation Institute for Life Sciences and Energy (MILES), HKUST-SUSTech, Shenzhen 518055, China (e-mail: ffwang@eee.hku.hk).
Pengxin Guo and Runxi Wang contributed equally to this work.
Corresponding author: Liangqiong Qu.

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그래디언트 역전 공격



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논문 분석



최적화 기반 공격
OP - GIA



생성 기반 공격
GEN - GIA



분석 기반 공격
ANA - GIA

논문 내용

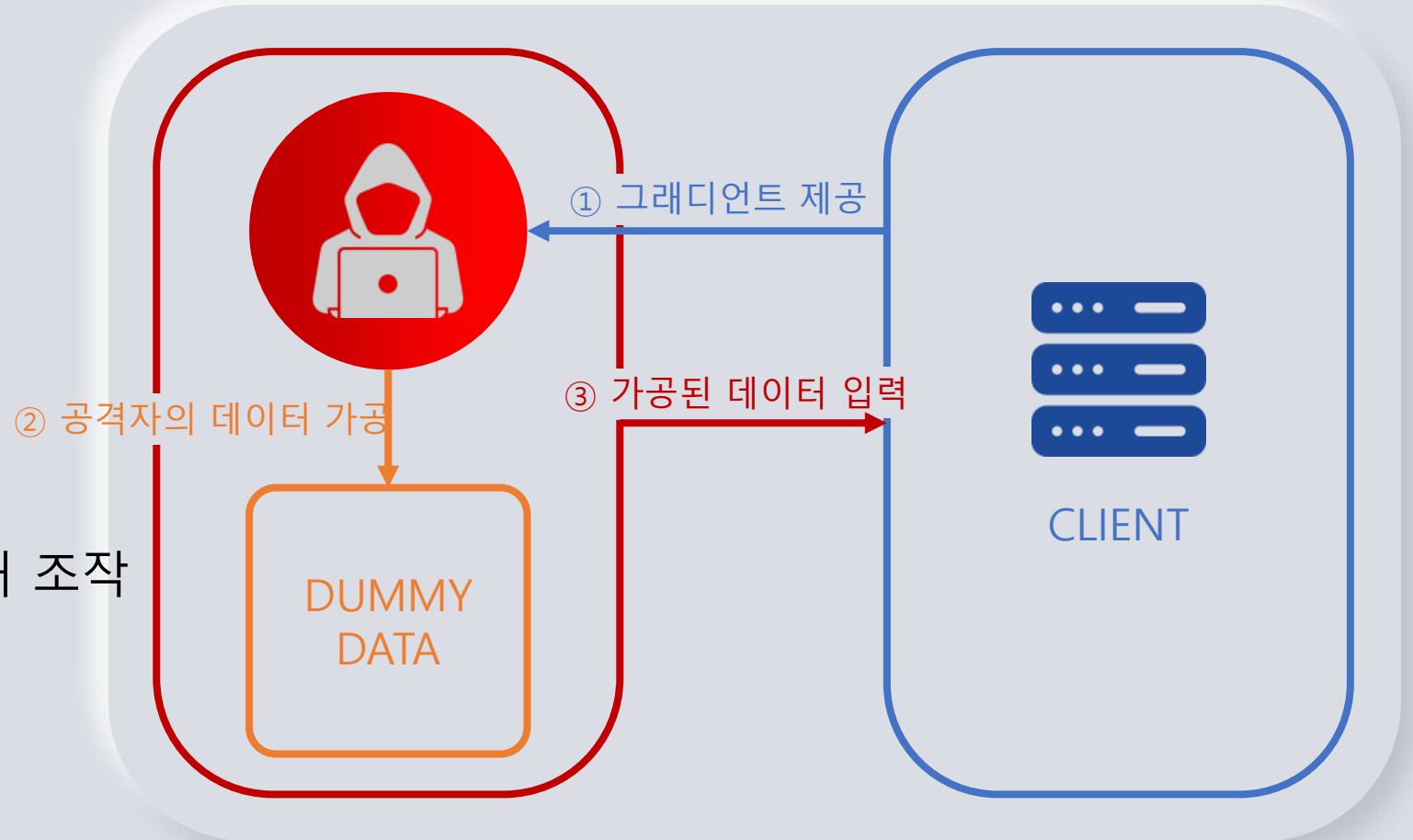
최적화 기반 공격

OP - GIA

Optimization-based
Gradient Inversion Attacks

핵심 특징

기울기 정보 기반의 반복적 데이터 조작
공격자에 의한 기울기 기반
데이터 반복 수정



최적화 기반 공격

논문 속 실험

데이터 레벨 실험

- 배치 크기
- 이미지 해상도
- 동일 레이블 데이터 수

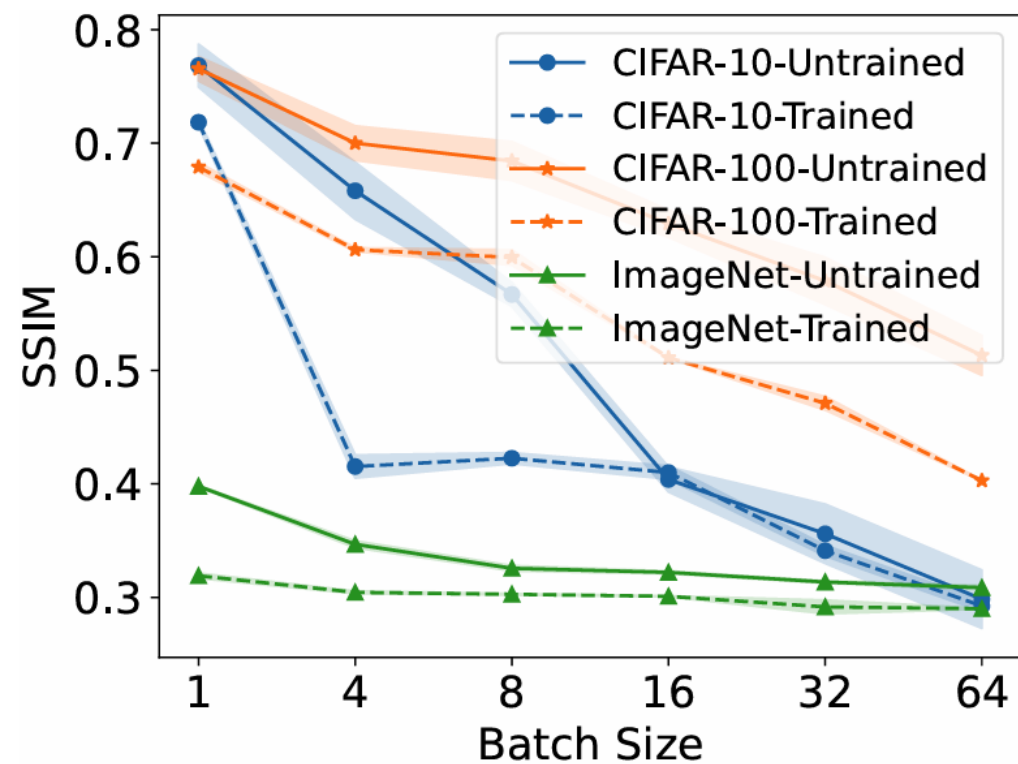
모델 레벨 실험

- 네트워크 아키텍처
- 모델 학습 상태

학습 프로토콜 실험

- 실제 연합 학습 환경 (FedAvg)에서의 실험

최적화 기반 공격 실험 결과



배치 크기 **증가**

공격 성능 **감소**

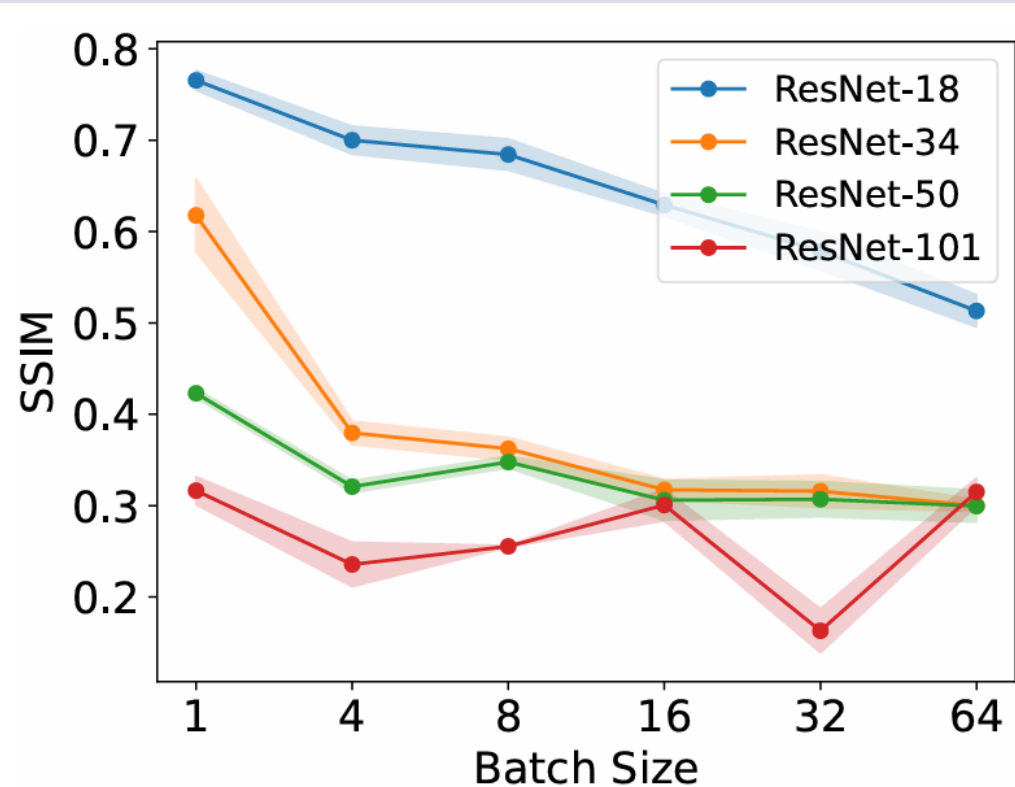
이미지 해상도 **증가**

공격 성능 **감소**

학습된 모델이면

공격 성능 **감소**

최적화 기반 공격 실험 결과



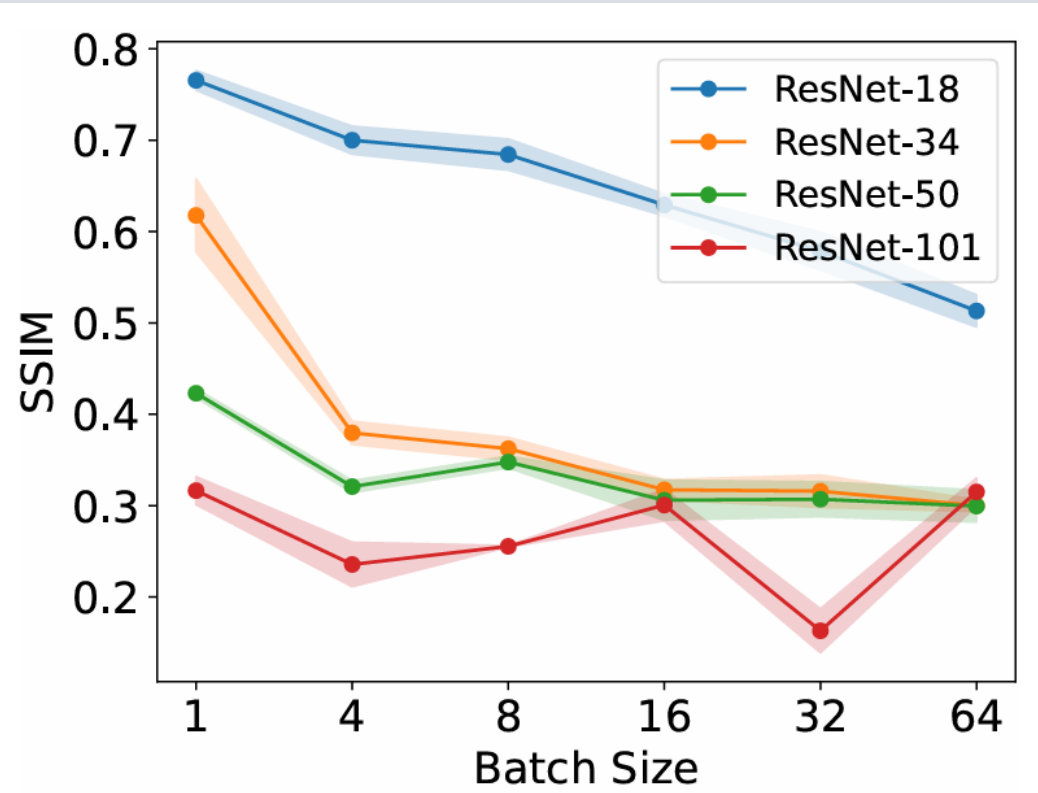
배치 크기 **증가**

공격 성능 **감소**

모델 복잡도 **증가**

공격 성능 **감소**

최적화 기반 공격 실험 결과



배치 크기 **증가** 공격 성능 **감소**

모델 복잡도 **증가** 공격 성능 **감소**

최적화 기반 공격 실험 결과

E	B	Strong Simulation			Weak Simulation			No Simulation		
		PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
1	1	12.02	0.3031	0.2049	12.26	0.3971	0.1522	11.40	0.2301	0.2252
1	8	14.11	0.5026	0.1295	11.75	0.3615	0.1681	11.53	0.1824	0.2512
1	16	14.50	0.5050	0.1315	12.26	0.3971	0.1522	11.57	0.1503	0.2751
2	1	12.13	0.3653	0.1701	11.08	0.2672	0.2120	11.30	0.2011	0.2372
2	8	12.99	0.4252	0.1486	12.36	0.3871	0.1624	11.44	0.1573	0.2877
2	16	15.17	0.5261	0.1138	13.67	0.4502	0.1409	11.62	0.1375	0.2880
5	1	11.51	0.3462	0.1816	10.93	0.2647	0.2363	11.48	0.1898	0.2482
5	8	14.58	0.5210	0.1232	11.63	0.3321	0.1928	10.48	0.1666	0.3045
5	16	14.85	0.5253	0.1137	12.22	0.3486	0.1702	11.03	0.1311	0.3281

FedAvg 자체가
OP-GIA 에 저항

최적화 기반 공격

성능 결정 요인



논문 내용
논문 분석



최적화 기반 공격
OP - GIA



생성 기반 공격
GEN - GIA



분석 기반 공격
ANA - GIA

논문 내용

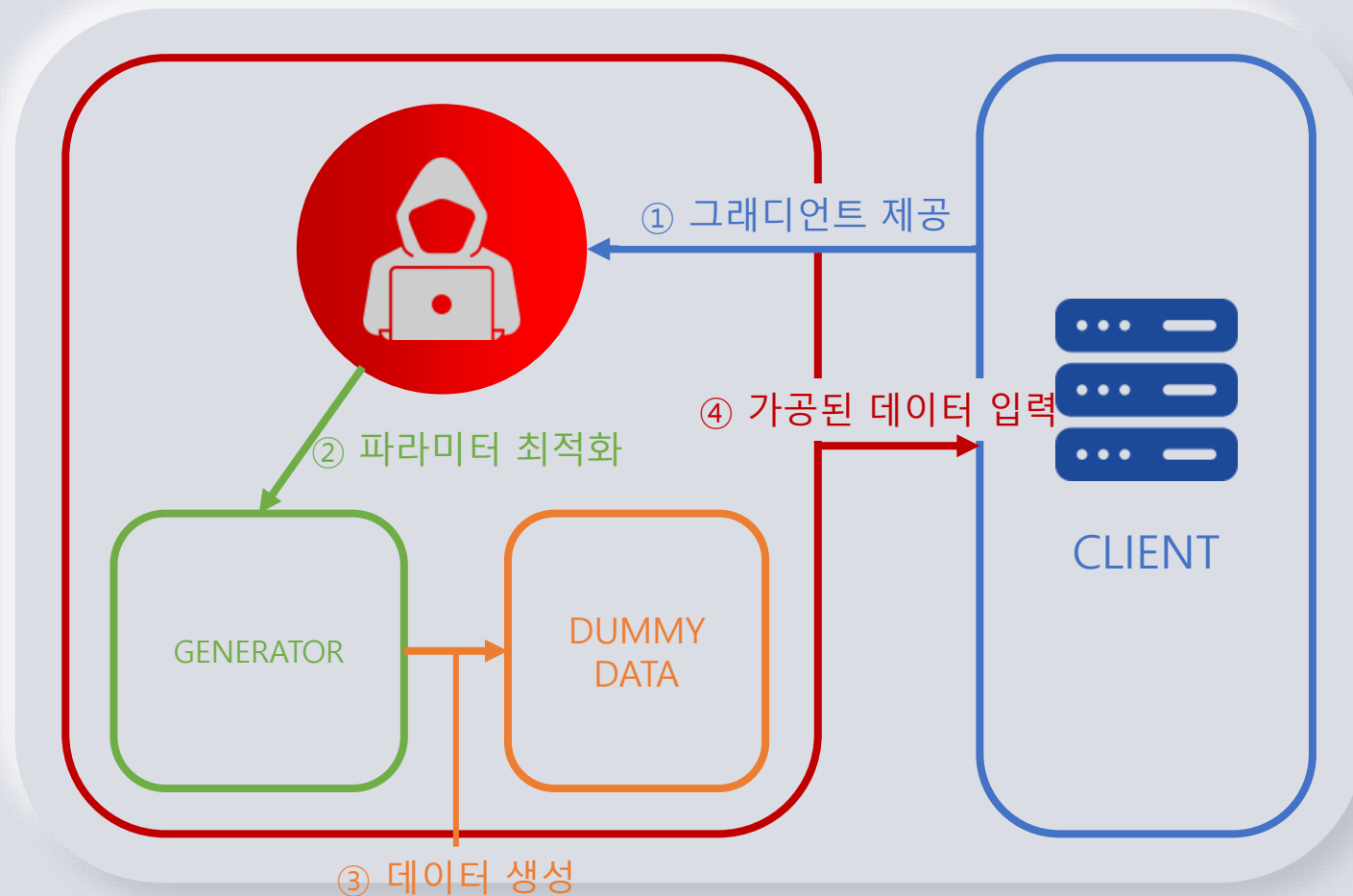
생성 기반 공격

GEN - GIA

Generation-based
Gradient Inversion Attacks

핵심 특징

기울기 기반 생성자 최적화 및
데이터 생성 기울기 정보를 활용한
생성자 학습과 데이터 생성



생성 기반 공격 논문 속 실험

잠재 벡터 최적화 실험

- 무작위 노이즈 실험
- 데이터 분포 불일치 실험

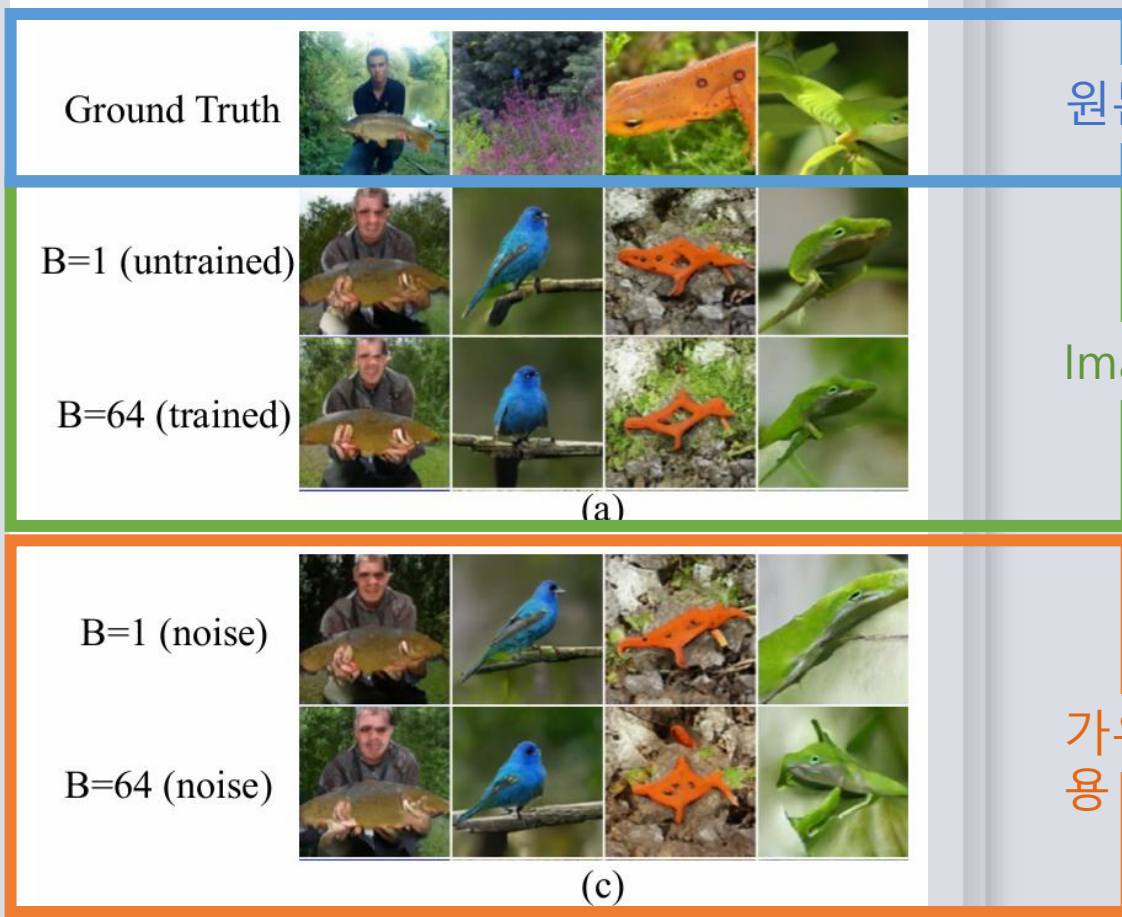
생성자 파라미터 최적화 실험

- 활성화 함수 비교 실험

역전 생성 모델 학습 실험

- OP-GIA 영향 요인과 비교 실험
- 보조 데이터셋 의존성 실험

생성 기반 공격 실험 결과



원본 데이터

ImageNet 재구성

가우시안 노이즈 적용

GEN-GIA의 의존도

그래디언트 정보



사전 학습된 생성자 성능
레이블 정보

생성 기반 공격 실험 결과

원본 데이터 (CIFAR-100)

Ground Truth



B=1



생성 데이터

생성자가 ImageNet 환경에서 사전학습



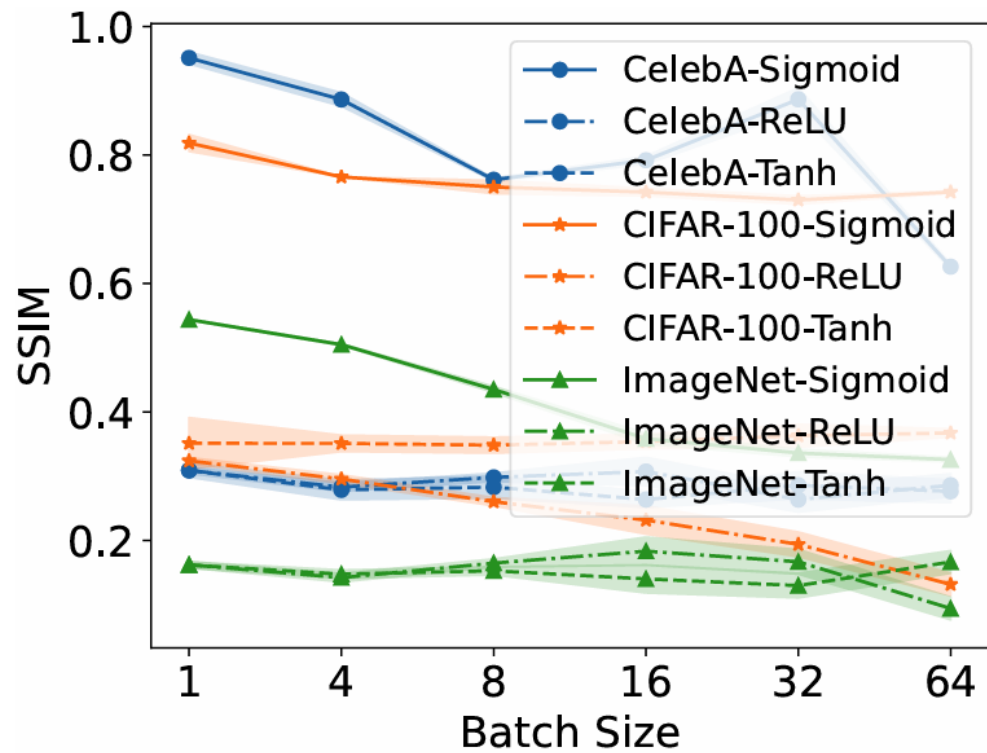
데이터셋 불일치



재구성 성능이 떨어짐

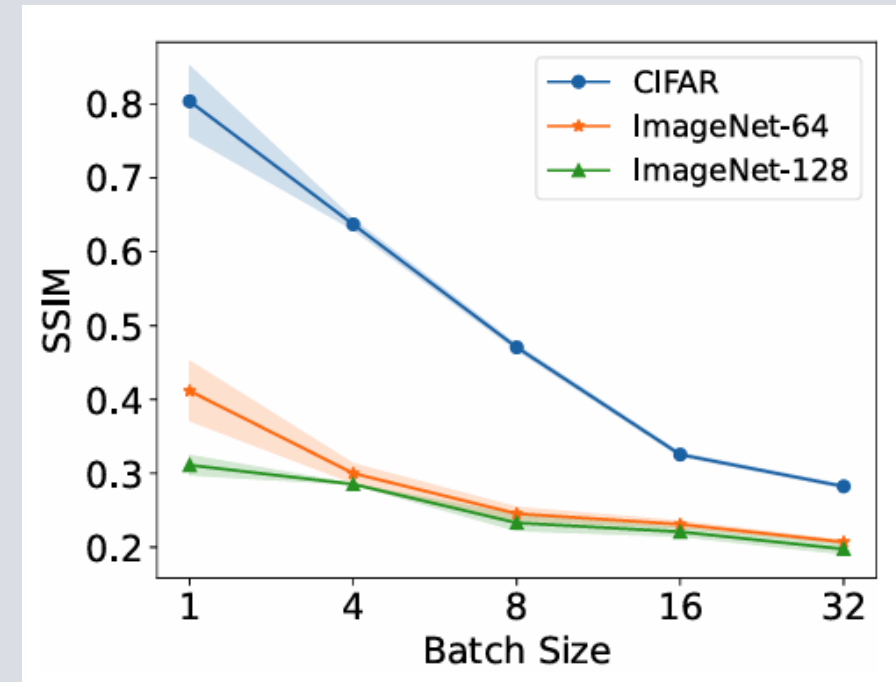
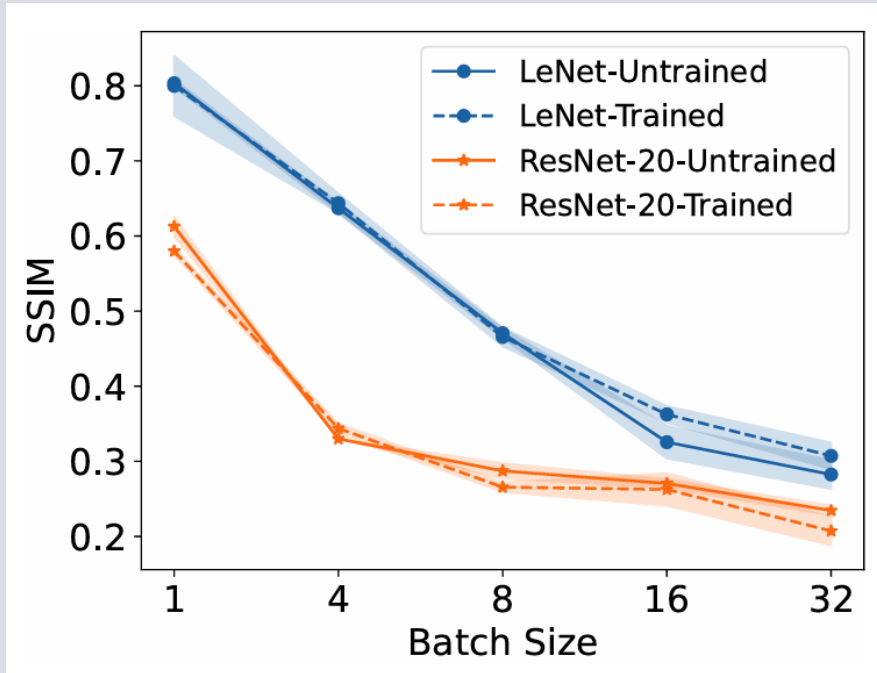
사전 학습된 생성자에 크게 의존

생성 기반 공격 실험 결과



활성화 함수	공격 성공 여부
Sigmoid	O
ReLU	X
Tanh	X

생성 기반 공격 실험 결과



모델의 학습 상태를 제외한 대부분 요인들이 성능 저하 발생 (OP-GIA와 유사)

생성 기반 공격

성능 결정 요인



논문 내용
논문 분석



최적화 기반 공격
OP - GIA



생성 기반 공격
GEN - GIA



분석 기반 공격
ANA - GIA

논문 내용

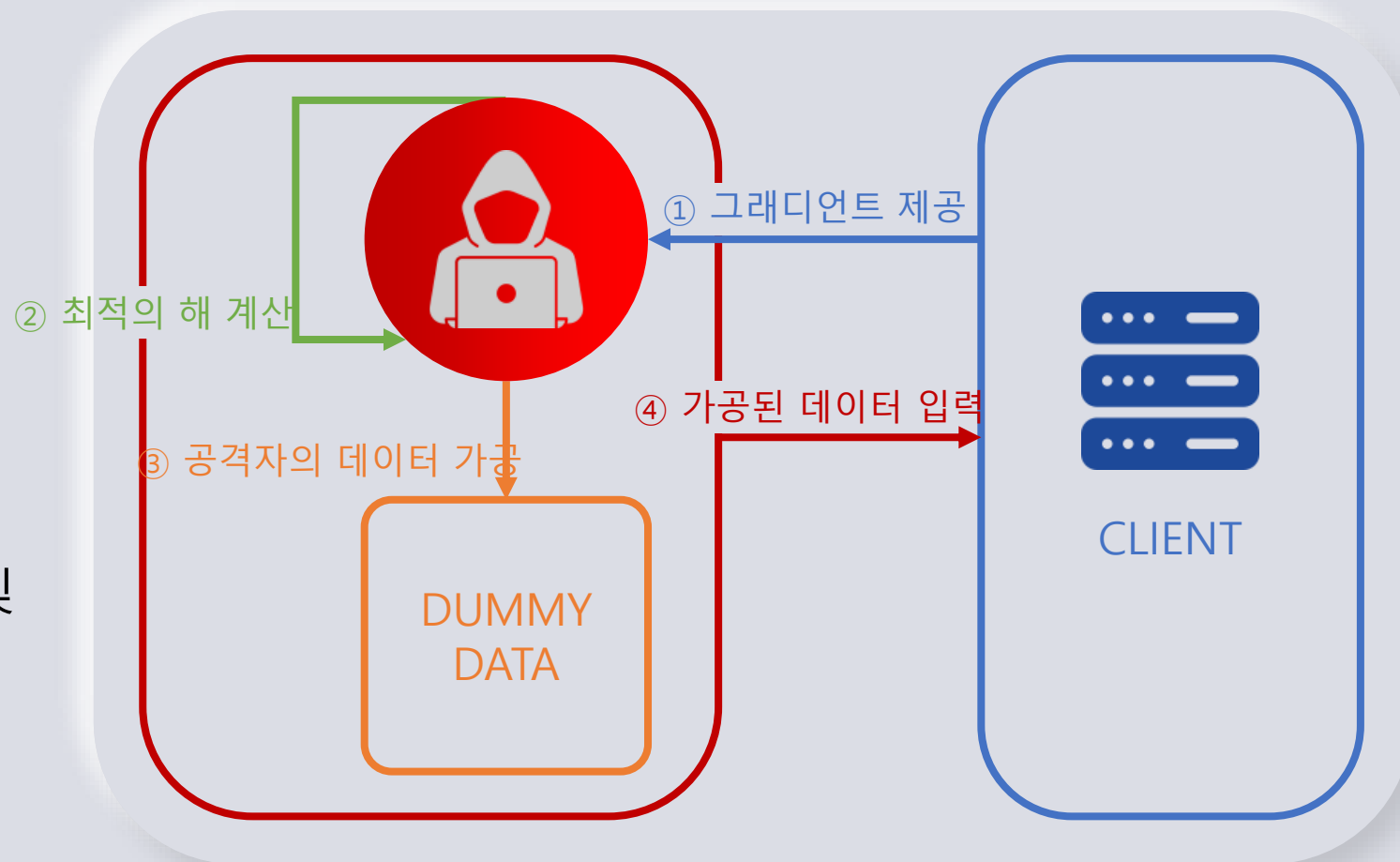
분석 기반 공격

ANA - GIA

Analytics-based
Gradient Inversion Attacks

핵심 특징

기울기 정보 기반의 완전해 도출 및
데이터 생성 기울기 정보를 활용한
수식적 최적해 탐색과 데이터 생성



분석 기반 공격

논문 속 실험

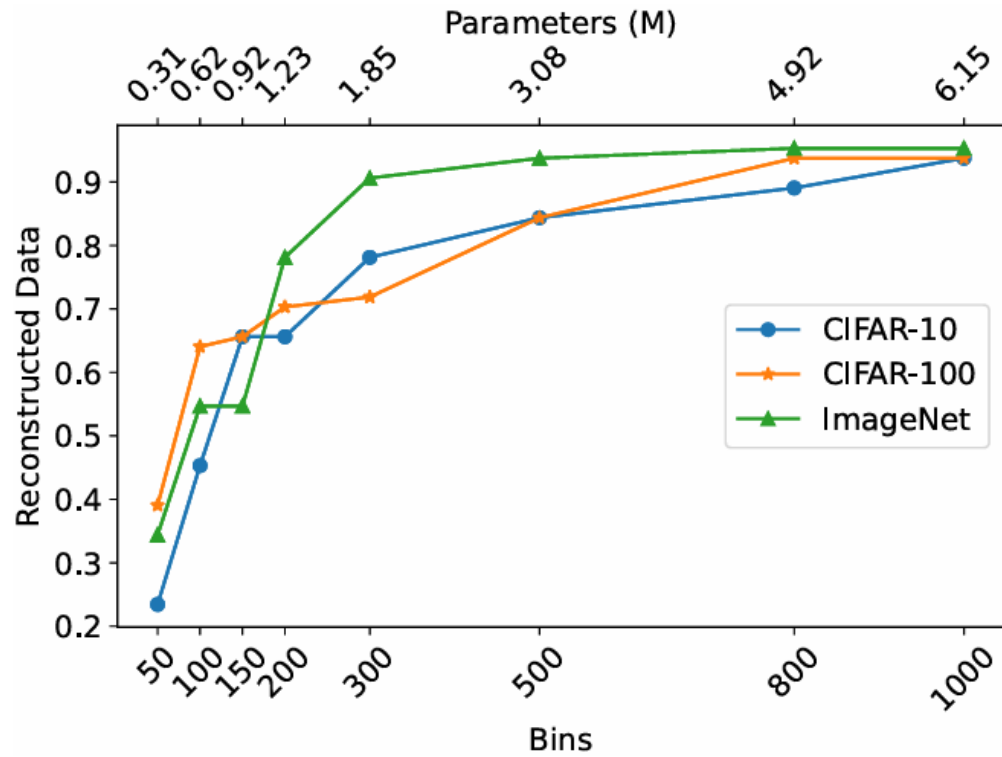
모델 구조 조작 실험

- 완벽한 복원 검증 실험
- 오버헤드 및 트레이드오프 분석

모델 파라미터 조작 실험

- 배치 크기 영향력 실험
- 해상도 및 모델 복잡도 실험
- 학습 상태의 역설

분석 기반 공격 실험 결과

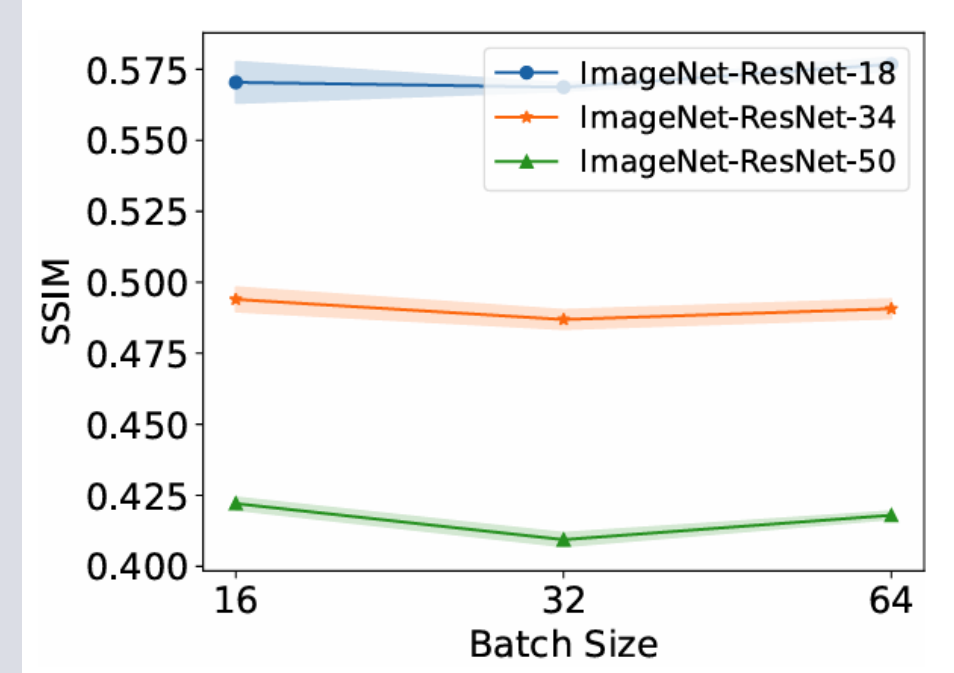
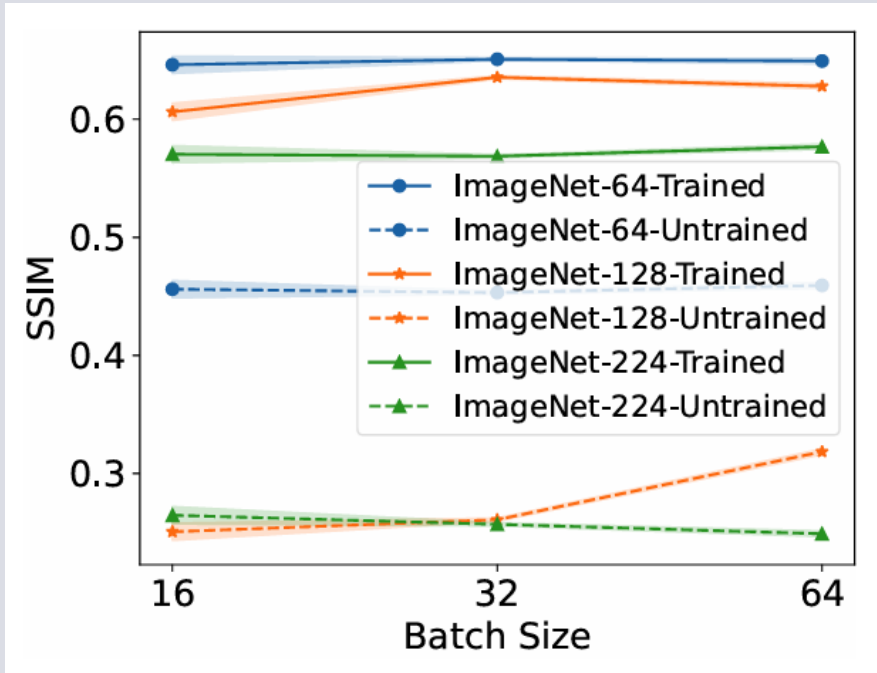


*빈(Bin)의 개수가 충분히 크다면
다른 외부 요인과 상관없이
모든 이미지 완벽하게 복원 가능

빈 개수 증가 → 파라미터 기하급수
적 증가

빈(Bin) : 임계값

분석 기반 공격 실험 결과



배치 크기와의 관계를 제외한 대부분 요인들이 성능 저하 발생 (OP-GIA와 유사)

분석 기반 공격

성능 결정 요인



논문 내용

방어 파이프라인

네트워크 설계 단계

Sigmoid 피하기
복잡한 네트워크 아키텍처

학습 프로토콜 단계

큰 배치 크기
여러 단계 로컬 학습 수행

클라이언트 검증 단계

파라미터 검증 절차 도입

실험 및 검증

실험 및 검증 대상

OP - GIA



배치 크기

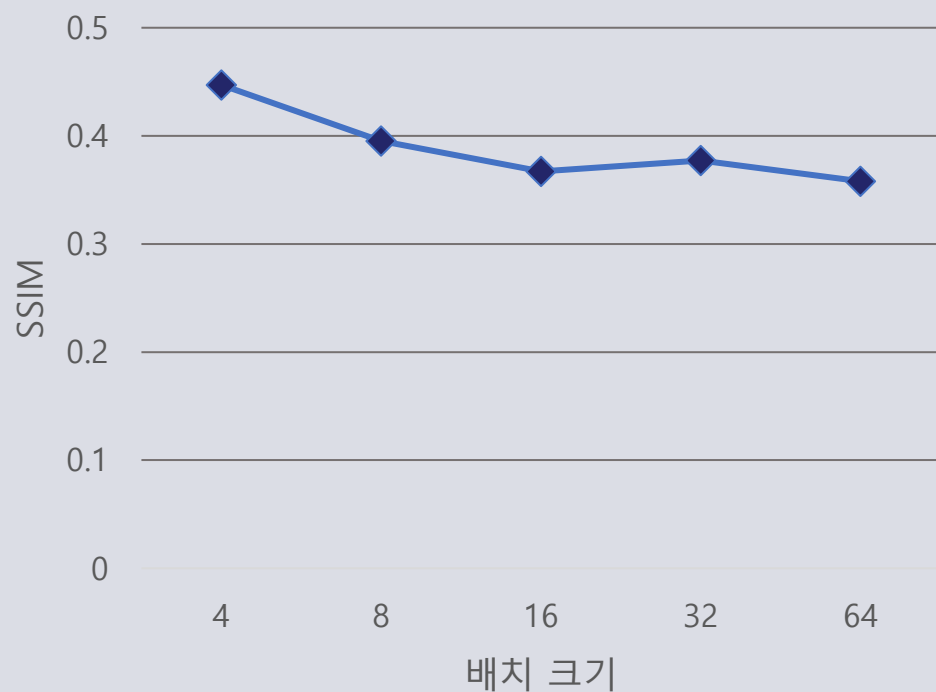
ANA - GIA



빈 개수

실험 및 검증

OP-GIA – 배치 크기



테스트 데이터셋: ImageNet-100

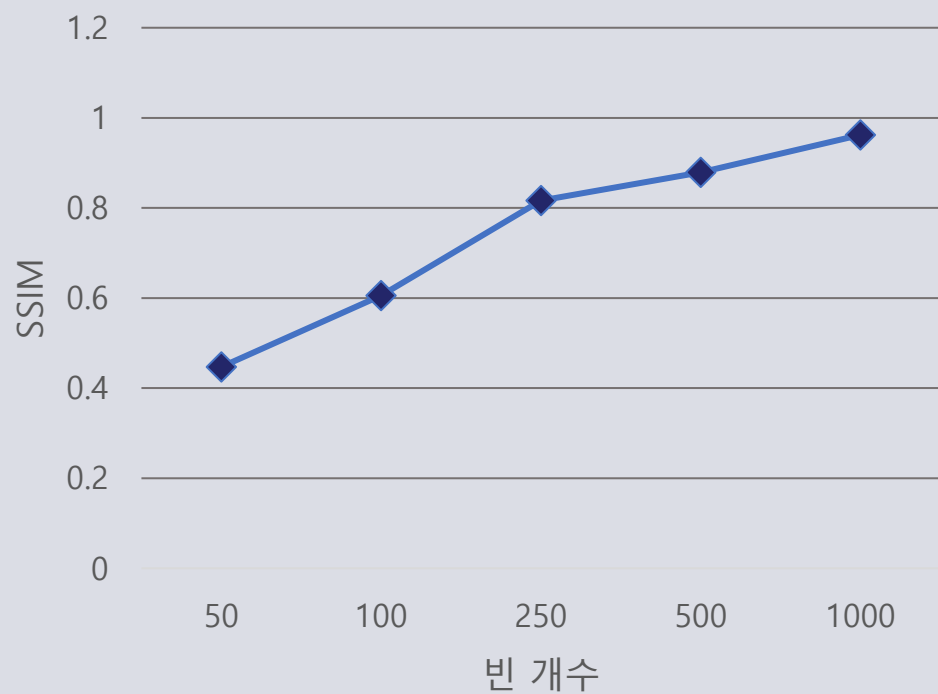
최적화 횟수: 3000

배치 크기	SSIM (구조적 유사도)
4	0.4471
8	0.3952
16	0.3671
32	0.3773
64	0.3580

논문과 동일하게 배치 크기가 커질 수록 성능 저하

실험 및 검증

ANA-GIA – 빈 개수



빈 개수	SSIM	원본 복원 능력
50	0.4512	복원 불가
100	0.6058	저품질
250	0.8162	고품질
500	0.8788	고품질
1000	0.9621	최고 품질

논문과 동일하게 빈 개수가 커질 수록 성능 향상

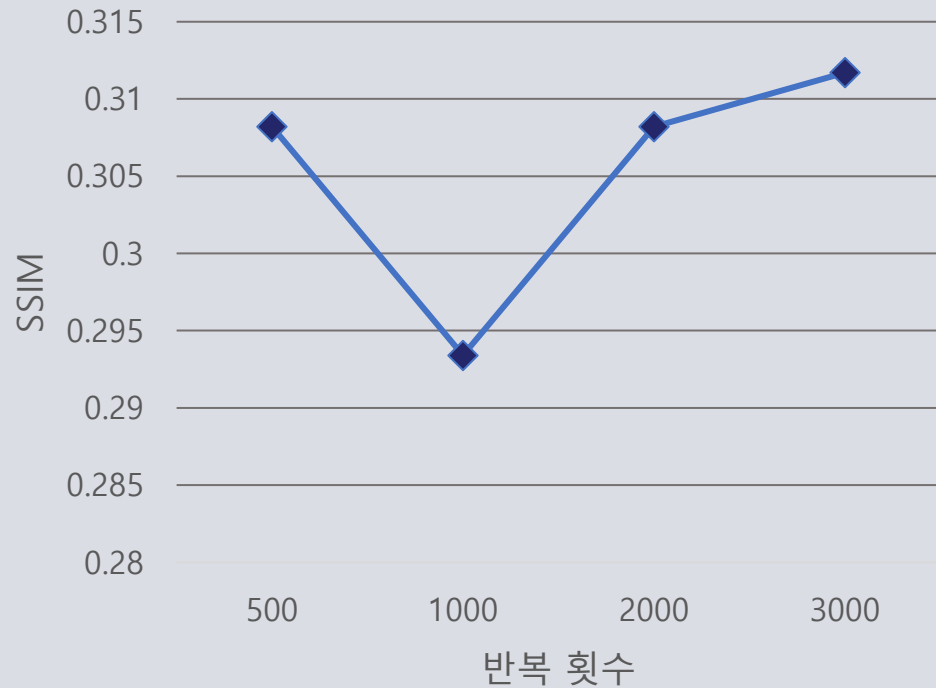
심화 탐구
심화 탐구

OP-GIA 새로운
파라미터 시도

방어 논문 탐색

심화 탐구

새로운 파라미터 시도



타겟 공격 기법: OP-GIA

최적화 반복 횟수 제외 모든 요소 동일

반복 횟수	SSIM (구조적 유사도)
500	0.3082
1000	0.2934
2000	0.3082
3000	0.3117

최적화 횟수가 바뀌어도 성능 변화가 없음

심화 탐구 방어 논문 탐색

arXiv:2412.07187v1 [cs.LG] 10 Dec 2024

A New Federated Learning Framework Against Gradient Inversion Attacks

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²Department of Mathematics, The University of Hong Kong

³College of Computer Science, Beijing University of Technology

⁴School of Mechanical Engineering and Automation, Harbin Institute of Technology, Shenzhen

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Abstract

Federated Learning (FL) aims to protect data privacy by enabling clients to collectively train machine learning models without sharing their raw data. However, recent studies demonstrate that information exchanged during FL is subject to Gradient Inversion Attacks (GIA) and, consequently, a variety of privacy-preserving methods have been integrated into FL to thwart such attacks, such as Secure Multi-party Computing (SMC), Homomorphic Encryption (HE), and Differential Privacy (DP). Despite their ability to protect data privacy, these approaches inherently involve substantial privacy-utility trade-offs. By revisiting the key to privacy exposure in FL under GIA, which lies in the frequent sharing of model gradients that contain private data, we take a new perspective by designing a novel privacy preserve FL framework that effectively “breaks the direct connection” between the shared parameters and the local private data to defend against GIA. Specifically, we propose a Hypernetwork Federated Learning (HyperFL) framework that utilizes hypernetworks to generate the parameters of the local model and only the hypernetwork parameters are uploaded to the server for aggregation. Theoretical analyses demonstrate the convergence rate of the proposed HyperFL, while extensive experimental results show the privacy-preserving capability and comparable performance of HyperFL. Code is available at <https://github.com/Pengxin-Guo/HyperFL>.

1 Introduction

Deep Neural Networks (DNN) have achieved remarkable success on a variety of computer vision tasks, relying on the availability of a large amount of training data (Krizhevsky, Sutskever, and Hinton 2012; He et al. 2016; Dosovitskiy et al. 2021; Liu et al. 2021; Wang et al. 2023). However, in many real-world applications, training data is distributed across different institutions, and data sharing between these entities is often restricted due to privacy and regulatory concerns. To alleviate these concerns, Federated Learning (FL) (McMahan et al. 2017; Li et al. 2020a; Qu et al. 2022; Zeng

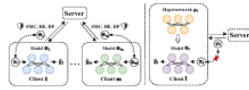


Figure 1: *Left*: Existing methods mainly explore defenses mechanisms on the shared gradients. Such mechanisms, including SMC, HE, and DP, inherently involve substantial privacy-utility trade-offs. *Right*: A novel FL framework that “breaks the direct connection” between the shared parameters and the local private data is proposed to achieve a favorable privacy-utility trade-off.

et al. 2024; Guo et al. 2024; Zhang et al. 2024) has emerged as a promising approach that enables collaborative and decentralized training of AI models across multiple institutions without sharing of personal data externally.

Despite the privacy-preserving capability introduced by FL, recent works (Geiping et al. 2020; Huang et al. 2021; Hatamizadeh et al. 2023) have revealed that FL models are vulnerable to Gradient Inversion Attacks (GIA) (Fredrikson, Jha, and Ristenpart 2015; Zhu, Liu, and Han 2019). GIA can reconstruct clients’ private data from the shared gradients, undermining FL’s privacy guarantees. To remedy this issue, various defense mechanisms have been integrated into FL, including Secure Multi-party Computing (SMC) (Yao 1982; Bonawitz et al. 2017; Mugunthan et al. 2019; Mou et al. 2021; Xu et al. 2022), Homomorphic Encryption (HE) (Gentry 2009; Zhang et al. 2020a,b; Ma et al. 2022; Park and Lim 2022) and Differential Privacy (DP) (Dwork 2006; Geyer, Klein, and Nabi 2017; McMahan et al. 2018; Yu, Bagdasaryan, and Shmatikov 2020; Bietti et al. 2022; Shen et al. 2023). These approaches primarily rely on existing defense mechanisms to enhance privacy protection against GIA without altering the FL framework, as illustrated in the left part of Figure 1. However, such mechanisms, including SMC, HE, and DP, inherently involve substantial privacy-utility trade-offs. For example, SMC and HE, while providing strong security guarantees through encrypted informa-

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